UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 2

Fall 2025

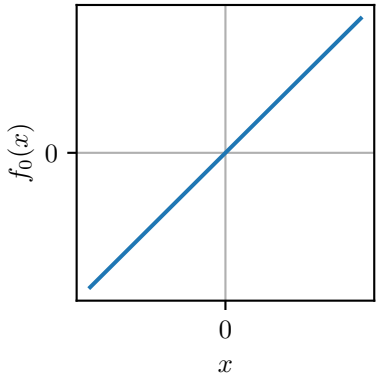
Instructor: Somayeh Sojoudi

Typeset by: George Ma

© EECS 127 - EECS 227AT. Optimization Models in Engineering

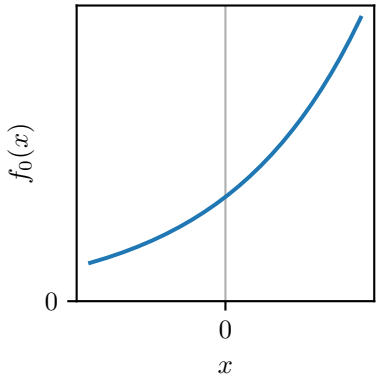
When we solve an optimization problem, we obtain $\boldsymbol{x}^*$ and $f_0(\boldsymbol{x}^*)$. Here, $f_0(\boldsymbol{x}^*)$ is called an optimal objective value, and $\boldsymbol{x}^*$ is called a minimizer in some textbooks.
Sometimes $\boldsymbol{x}^*$ does not exist:

1. $\min x$:



   $x^* = -\infty$ is not a number.

2. $\min \mathrm{e}^x$:



   $x^* = -\infty$ is not a valid choice.

- Case 1: solution x^* does not exist and the problem is unbounded from below.
- Case 2: solution x^* is not attainable, but the optimal objective value exists.

Definition 1 (Infeasible problem)**.** If a problem has no feasible solution, it is called an *infeasible problem.*

Example:

$$\begin{aligned} \min \quad & x, \\ \text{s.t.} \quad & \underbrace{\boxed{x^2 \le -1}}_{\text{no } x \text{ satisfies this constraint}}. \end{aligned}$$

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.

In that case, by convention, we say the optimal objective value is $+\infty$.

$$\text{Optimal objective value} \begin{cases} +\infty: & \text{infeasible,} \\ \text{finite}: & n^* \text{ may or may not be attainable,} \\ -\infty: & \text{unbounded from below.} \end{cases}$$

Optimization problems can be tractable or not tractable:

- Tractable: there is an algorithm to solve it efficiently.
- Not tractable: there is no algorithm to solve it fast enough (see example below).

Example 1. We want to make n decisions for a company to maximize the profit. Each decision is yes or no. Mathematically:

- We map each decision x_i $(i = 1, \ldots, n)$ to $+1$ (yes) and -1 (no).
- Let the profit as a function of decisions be written as $f_0(\boldsymbol{x})$.

Optimization:

$$\begin{aligned} \max \quad & f_0(\boldsymbol{x}) \iff \min \ -f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & x_i \in \{-1, 1\}, \ i = 1, \ldots, n \\ \iff & x_i^2 = 1 \\ \iff & \text{standard form: } \begin{cases} x_i^2 - 1 \le 0, \\ -x_i^2 + 1 \le 0. \end{cases} \end{aligned}$$

Number of feasible points: 2^n (each feasible solution is $[\pm 1, \ldots, \pm 1]^{\mathrm{T}}$).
For a general $f_0(\boldsymbol{x})$, there is no better algorithm than simply checking the objective value at all feasible points to get the best one.

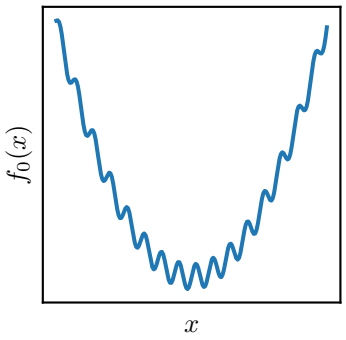
$$\text{number of function evaluations/checkings} = 2^n.$$

If $n = 500$, the number of required function evaluations is greater than the number of atoms in the observable universe. This is extremely slow and it is called *intractable*.
(2^n is exponential and is called intractable. A polynomial function of n for complexity is called tractable.) ■

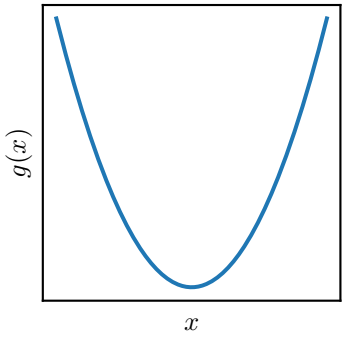
This class focuses on convex optimization that is *tractable*.
The above problem, named *combinatorial optimization*, is intractable.
In other classes, you will learn how to solve an intractable problem *approximately* via one or a sequence of tractable problems (solved approximately because solving it precisely takes so much time).

Example 2. Find the minimum of $f_0(x)$:

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.



$f_0(x)$ has so many local minima and finding the global minimum is hard. However, we can approximate it with an easier problem:



We need to find a good $g(x)$ whose global minimum is close to the global minimum of $f_0(x)$.

- $f_0(x)$ is not convex while $g(x)$ is convex.
- Focus of this class is on different types of convex optimization, such as least-squares, linear program, quadratic program and conic optimization, with different applications to machine learning and artificial intelligence.

■

Review of linear algebra. *Space*: collections of objects of a certain type.
Example:

1. $\mathbb{R}^n$: collection of vectors with n elements.
2. Collection of polynomials with one variable such as $a_0 + a_1 t + a_2 t^2 + \cdots + a_k t^k$, where t is the argument of the polynomial.

In this class, we mostly focus on the $\mathbb{R}^n$ space. (Later on we also consider the space of matrices.)

Definition 2 (Subspace)**.** A non-empty set V is called a *subspace* of space $\mathbb{R}^n$ if

$$\forall \boldsymbol{x}, \boldsymbol{y} \in V \Longrightarrow \alpha \boldsymbol{x} + \beta \boldsymbol{y} \in V \text{ for all scalars } \alpha, \beta.$$

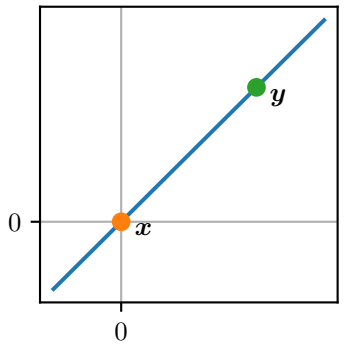
In other words, the linear combination of every two vectors of V is again in V.

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.

In Definition 2, by setting $\alpha = \beta = 0$, we have $\mathbf{0} \in V$. Therefore, a subspace contains the origin.

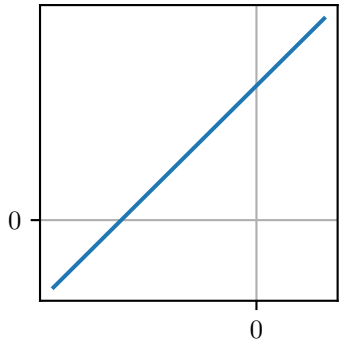
Example 3. Example: consider $\mathbb{R}^2$.

- $\mathbb{R}^2$ is a subspace of $\mathbb{R}^2$.
- A line passing through the origin is a subspace of $\mathbb{R}^2$.



  If $\boldsymbol{x}, \boldsymbol{y}$ belong to the line, then $\alpha\boldsymbol{x} + \beta\boldsymbol{y}$ also belongs to the line.

- A line not passing through the origin is *not* a subspace (because if $\boldsymbol{x}, \boldsymbol{y}$ belong to the line, $0\boldsymbol{x} + 0\boldsymbol{y} = \mathbf{0}$ does not belong to the line).



■

Consider m vectors $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(m)} \in \mathbb{R}^n$. Define $\text{span}(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(m)})$ as the set of all vectors that can be written as a linear combination of $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(m)}$.
Note: $\text{span}(\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)})$ is a subspace.

Definition 3 (Basis)**.** Consider a subspace V. A set of vectors $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(d)} \in V$ is called a *basis* of V if

1. $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(d)}$ are linearly independent.
2. For every $\boldsymbol{x} \in V$, there exist coefficients $\alpha_1, \ldots, \alpha_d$ such that $\boldsymbol{x} = \alpha_1\boldsymbol{x}^{(1)} + \cdots + \alpha_d\boldsymbol{x}^{(d)}$.

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.

Recall: $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(d)}$ are called *linearly independent* if there does not exist a set of coefficients $\beta_1, \ldots, \beta_d \in \mathbb{R}$ such that

1. At least one of $\beta_1, \ldots, \beta_d$ is non-zero.
2. $\beta_1 \boldsymbol{x}^{(1)} + \cdots + \beta_d \boldsymbol{x}^{(d)} = \mathbf{0}$.

In other words, the only linear combination of $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(d)}$ that produces the zero vector is by zero coefficients.
Given a subspace V, the set of basis is *not* unique. But all basis sets have the same number of vectors d. Here, d is called the *dimension* of the subspace V.

Example 4. Consider

$$\boldsymbol{x}^{(1)} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{x}^{(2)} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \boldsymbol{x}^{(3)} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Define $V = \text{span}(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \boldsymbol{x}^{(3)})$.
Note that $(-2)\boldsymbol{x}^{(1)} + (+1)\boldsymbol{x}^{(2)} + (-1)\boldsymbol{x}^{(3)} = \mathbf{0}$, thus $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \boldsymbol{x}^{(3)}$ are linearly dependent.
Also, if $\beta_1 \boldsymbol{x}^{(1)} + \beta_2 \boldsymbol{x}^{(2)} = \mathbf{0}$, then

$$\begin{cases} \beta_1 + \beta_2 = 0, \\ \beta_1 + 2\beta_2 = 0, \\ \beta_1 + 3\beta_2 = 0 \end{cases} \implies \beta_1 = \beta_2 = 0.$$

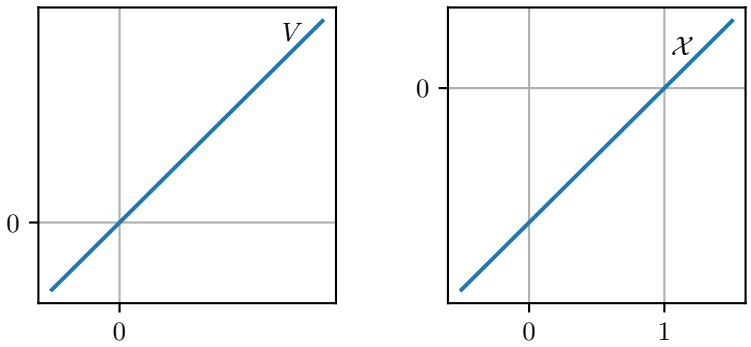
Thus, $\boldsymbol{x}^{(1)}$ and $\boldsymbol{x}^{(2)}$ are linearly independent.
$\{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}\}$ is a basis for V and $\dim(V) = 2$. Here dim stands for dimension. ■

Definition 4 (Affine set)**.** A set $\mathcal{X} \in \mathbb{R}^n$ is called *affine* if there is a subspace V of $\mathbb{R}^n$ and a vector $\boldsymbol{x}^{(0)}$ in $\mathbb{R}^n$ such that

$$\mathcal{X} = \boldsymbol{x}^{(0)} + V,$$

meaning that we add vector $\boldsymbol{x}^{(0)}$ to all vectors in V to obtain $\mathcal{X}$.

Example 5. $\mathcal{X} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + V$:



© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.

■

Affine set is a shifted version of subspace and may not pass through the origin.
So far, we focused on space of vectors in $\mathbb{R}^n$. Later on, we also study space of matrices in $\mathbb{R}^{n\times m}$.
We define *norm* in $\mathbb{R}^n$. It is similar for $\mathbb{R}^{n\times m}$.

Definition 5 (Norm)**.** A function $\|\cdot\|\colon \mathbb{R}^n \to \mathbb{R}$ is a *norm* if

1. $\|\boldsymbol{x}\| \geq 0,\ \forall \boldsymbol{x} \in \mathbb{R}^n$ and $\|\boldsymbol{x}\| = 0$ iff $\boldsymbol{x} = \mathbf{0}$.
2. $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|,\ \forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$.
3. $\|\alpha\boldsymbol{x}\| = |\alpha|\|\boldsymbol{x}\|,\ \forall \alpha \in \mathbb{R}, \boldsymbol{x} \in \mathbb{R}^n$.

Examples of norm:

- L_1-norm: $\|\boldsymbol{x}\|_1 = |x_1| + \cdots + |x_n|$.
- L_2-norm: $\|\boldsymbol{x}\|_2 = \sqrt{x_1^2 + \cdots + x_n^2}$.
- L_∞-norm: $\|\boldsymbol{x}\|_\infty = \max\{|x_1|, \ldots, |x_n|\}$.
- General L_p-norm for $1 \leq p < \infty$:

$$\|\boldsymbol{x}\|_p = \left(|x_1|^p + \cdots + |x_n|^p\right)^{\frac{1}{p}}.$$

(L in L_p stands for length as it shows length of $\boldsymbol{x}$ with respect to some measures.)

Note: $\|\boldsymbol{x}\|_p$ is a norm if $p \geq 1$ and is not a norm if $p < 1$.
Define

$$\begin{aligned}\|\boldsymbol{x}\|_0 &= \lim_{p\to 0} \left(|x_1|^p + \cdots + |x_n|^p\right)^{\frac{1}{p}} \\ &= \text{cardinality of } \boldsymbol{x} \\ &= \text{number of non-zero elements in } \boldsymbol{x}.\end{aligned}$$

$\|\boldsymbol{x}\|_0$ is not a norm, but appears frequently in machine learning.

Definition 6 (Inner product)**.** An *inner product* on a space X (such as $\mathbb{R}^n$ or $\mathbb{R}^{n\times m}$) is a function that takes two elements of X and maps them to a scalar. Inner product is written as $\langle\cdot\rangle\colon X \times X \to \mathbb{R}$ such that:

1. $\langle \boldsymbol{x}, \boldsymbol{x}\rangle \geq 0,\ \forall \boldsymbol{x} \in X$.
2. $\langle \boldsymbol{x}, \boldsymbol{x}\rangle = 0$ iff $\boldsymbol{x} = \mathbf{0}$.
3. $\langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z}\rangle = \langle \boldsymbol{x}, \boldsymbol{z}\rangle + \langle \boldsymbol{y}, \boldsymbol{z}\rangle,\ \forall \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in X$.
4. $\langle \alpha\boldsymbol{x}, \boldsymbol{y}\rangle = \alpha\langle \boldsymbol{x}, \boldsymbol{y}\rangle,\ \forall \boldsymbol{x}, \boldsymbol{y} \in X, \alpha \in \mathbb{R}$.
5. $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \langle \boldsymbol{y}, \boldsymbol{x}\rangle,\ \forall \boldsymbol{x}, \boldsymbol{y} \in X$.

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.

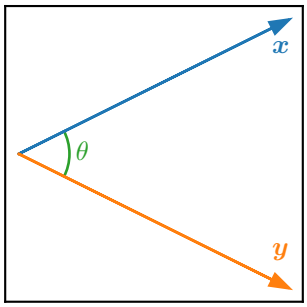
An example of inner product on $\mathbb{R}^n$ is

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \beta_1 x_1 y_1 + \beta_2 x_2 y_2 + \cdots + \beta_n x_n y_n,$$

where $\beta_1, \beta_2, \ldots, \beta_n$ are positive constants. If we pick $\beta_1 = \beta_2 = \cdots = \beta_n = 1$, we obtain the standard inner product that we use in this class:

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x}^{\mathrm{T}} \boldsymbol{y} = \boldsymbol{y}^{\mathrm{T}} \boldsymbol{x} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

Note that: $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \|\boldsymbol{x}\|_2 \cdot \|\boldsymbol{y}\|_2 \cdot \cos(\theta)$, where θ is the angle between vectors $\boldsymbol{x}$ and $\boldsymbol{y}$.



If $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$, then $\theta = 90^\circ$ and the two vectors are orthogonal.
Example: given two articles published by two news outlets CNN and Fox News, we want to study their similarity.

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.